

```
!wget https://raw.githubusercontent.com/mwaskom/seaborn-data/master/tips.csv
```

```
content.com/mwaskom/seaborn-data/master/tips.csv
```

```
usercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...  
hubusercontent.com)|185.199.108.133|:443... connected.
```

```
9.50K  --.-KB/s    in 0s
```

```
red [9729/9729]
```

```
from pyspark.sql import SparkSession

spark = SparkSession.builder.appName("BigDataAnalysis").getOrCreate()

df = spark.read.csv("tips.csv", header=True, inferSchema=True)

df.show(5)
df.printSchema()

print("Total Records:", df.count())

df.describe().show()
```

```
-----+-----+-----+-----+-----+
sex|smoker|day|  time|size|
-----+-----+-----+-----+-----+
female|    No|Sun|Dinner|    2|
Male|    No|Sun|Dinner|    3|
Male|    No|Sun|Dinner|    3|
Male|    No|Sun|Dinner|    2|
female|    No|Sun|Dinner|    4|
-----+-----+-----+-----+-----+
rows
```

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nullable (nullable = true)
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```

```
-----+-----+-----+-----+-----+-----+-----+
total_bill|          tip|  sex|smoker| day|  time|          size|
-----+-----+-----+-----+-----+-----+-----+
      244|          244|  244|  244| 244|  244|          244|
522950824|2.9982786885245902| NULL| NULL|NULL| NULL| 2.569672131147541|
354856857|1.3836381890011815| NULL| NULL|NULL| NULL|0.9510998047322347|
```

3.07	1.0	Female	No	Fri	Dinner	1
50.81	10.0	Male	Yes	Thur	Lunch	6
-----+-----+-----+-----+-----+-----+-----+						

```
!wget https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic
```

```

https://raw.githubusercontent.com/datascopeio/datasets/master/titanic.csv
tent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.111.133,
ercontent.com (raw.githubusercontent.com)|185.199.108.133|:443... connecte
g response... 200 OK
plain]

=====>] 58.89K  --.-KB/s      in 0.01s

MB/s) - 'titanic.csv' saved [60302/60302]

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```
=====>] 58.89K --.-KB/s in 0.01s
MB/s) - 'titanic.csv' saved [60302/60302]
```

```
MB/s) - 'titanic.csv' saved [60302/60302]
```

```
df = spark.read.csv("titanic.csv", header=True, inferSchema=True)

df.show(5)
df.describe().show()
```

```
df.show(5)
df.describe().show()
```

```
df.show(5)
df.describe().show()
```

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch
1	0	3	Braund, Mr. Owen ...	male	22.0	1	0
2	1	1	Cumings, Mrs. Joh...	female	38.0	1	0
3	1	3	Heikkinen, Miss. ...	female	26.0	0	0
4	1	1	Futrelle, Mrs. Ja...	female	35.0	1	0
5	0	3	Allen, Mr. Willia...	male	35.0	0	0

```
only showing top 5 rows
```

```

+-----+-----+-----+-----+
|summary| PassengerId| Survived| Pclass|
+-----+-----+-----+-----+
| count|      891|      891|      891|
|  mean|    446.0| 0.3838383838383838| 2.308641975308642|
| stddev|257.3538420152301|0.48659245426485753|0.8360712409770491|
|   min|         1|         0|         1|"Anders
|   max|      891|         1|         3|van Mel
+-----+-----+-----+-----+

```

```
# Total Passengers
print("Total Passengers:", df.count())

# Survivors Count
df.groupby("Survived").count().show()

# Gender Count
```

```
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df.groupBy("Survived").count().show()

# Gender Count
```

```
# Survivors Count
df.groupBy("Survived").count().show()

# Gender Count
```

#	Gender	Count
1	Male	1
2	Female	1
3	Male	1
4	Female	1
5	Male	1
6	Female	1
7	Male	1
8	Female	1
9	Male	1
10	Female	1
11	Male	1
12	Female	1
13	Male	1
14	Female	1
15	Male	1
16	Female	1
17	Male	1
18	Female	1
19	Male	1
20	Female	1
21	Male	1
22	Female	1
23	Male	1
24	Female	1
25	Male	1
26	Female	1
27	Male	1
28	Female	1
29	Male	1
30	Female	1
31	Male	1
32	Female	1
33	Male	1
34	Female	1
35	Male	1
36	Female	1
37	Male	1
38	Female	1
39	Male	1
40	Female	1
41	Male	1
42	Female	1
43	Male	1
44	Female	1
45	Male	1
46	Female	1
47	Male	1
48	Female	1
49	Male	1
50	Female	1
51	Male	1
52	Female	1
53	Male	1
54	Female	1
55	Male	1
56	Female	1
57	Male	1
58	Female	1
59	Male	1
60	Female	1
61	Male	1
62	Female	1
63	Male	1
64	Female	1
65	Male	1
66	Female	1
67	Male	1
68	Female	1
69	Male	1
70	Female	1
71	Male	1
72	Female	1
73	Male	1
74	Female	1
75	Male	1
76	Female	1
77	Male	1
78	Female	1
79	Male	1
80	Female	1
81	Male	1
82	Female	1
83	Male	1
84	Female	1
85	Male	1
86	Female	1
87	Male	1
88	Female	1
89	Male	1
90	Female	1
91	Male	1
92	Female	1
93	Male	1
94	Female	1
95	Male	1
96	Female	1
97	Male	1
98	Female	1
99	Male	1
100	Female	1

```
df.groupBy("Sex").count().show()

# Passenger Class Count
df.groupBy("Pclass").count().show()

# Average Fare
from pyspark.sql.functions import avg

df.select(avg("Fare")).show()
```

Total Passengers: 891

```
+-----+-----+
|Survived|count|
+-----+-----+
|      1|   342|
|      0|   549|
+-----+-----+
```

```
+-----+-----+
|   Sex|count|
+-----+-----+
|female|   314|
|  male|   577|
+-----+-----+
```

```
+-----+-----+
|Pclass|count|
+-----+-----+
|      1|   216|
|      3|   491|
|      2|   184|
+-----+-----+
```

```
+-----+
|      avg(Fare)|
+-----+
|32.2042079685746|
+-----+
```

Check Missing Values

```
from pyspark.sql.functions import col, when, count

df.select([
    count(when(col(c).isNull(), c)).alias(c)
    for c in df.columns
]).show()
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|PassengerId|Survived|Pclass|Name|Sex|Age|SibSp|Parch|Ticket|Fare|Cabin|E
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|          0|        0|        0|  0|  0|177|        0|        0|        0|    0|  687|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
```

```
# Average Age by Passenger Class

from pyspark.sql.functions import avg

df.groupBy("Pclass").agg(
    avg("Age").alias("Average_Age")
).show()
```

```
+-----+-----+
|Pclass|      Average_Age|
+-----+-----+
|      1|38.233440860215055|
|      3| 25.14061971830986|
|      2| 29.87763005780347|
+-----+-----+
```